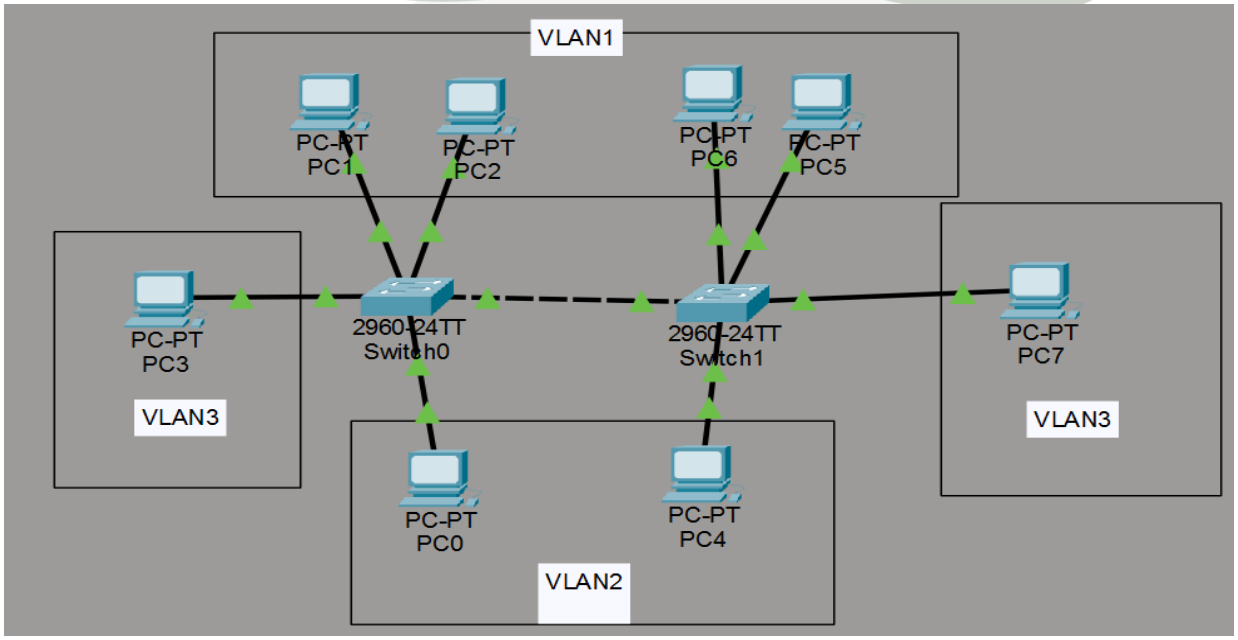
Aim: To configure VLANs on two switches and establish communication between PCs belonging to the same VLAN using Cisco Packet Tracer.

Description: Two Cisco 2960 switches are connected through a trunk link, and the PCs are divided into VLAN 1, VLAN 2 (CSE), and VLAN 3 (ECE). The required IP addresses are assigned to the PCs, and the switch ports connected to the PCs are configured as access ports and assigned to their respective VLANs. The link between the switches is configured as a trunk to carry VLAN 1, 2 and 3 traffic. The configuration is verified using show vlan brief and show interfaces trunk, and connectivity is tested using ping/PDU.

Procedure / Steps:

1. Start with **IP address configuration on all PCs**.
2. Configure the switches by creating the required VLANs using VLAN commands.
3. Configure the PC-connected ports as **access ports**.
4. Assign the required VLAN to each access port using switchport access vlan.
5. Configure the link between the two switches as a **trunk port** using switchport mode trunk.
6. Use switchport nonegotiate to prevent DTP negotiation.
7. Configure the VLANs allowed on the trunk.
8. Verify the complete configuration using the appropriate show commands.
9. Test connectivity between PCs using ping.

Network Topology



 Signature of the Faculty.....

IP Address Configuration

Switch	PC	IP Address	VLAN
Switch0	PC0	192.168.12.9	2
Switch0	PC1	192.168.12.10	1
Switch0	PC2	192.168.12.11	1
Switch0	PC3	192.168.12.12	3
Switch1	PC4	192.168.12.13	2
Switch1	PC5	192.168.12.14	1
Switch1	PC6	192.168.12.15	1
Switch1	**PC7**	**192.168.12.16**	**3**

Subnet Mask:255.255.255.0

Create VLANs

Configure the following VLANs on both switches.

Switch(config)# vlan 1

Switch(config-vlan)# exit

Switch(config)# vlan 2

Switch(config-vlan)# name CSE

Switch(config-vlan)# exit

Switch(config)# vlan 3

Switch(config-vlan)# name ECE

Switch(config-vlan)# exit

VLAN ID	VLAN Name
1	1
2	CSE
3	ECE

Assign Access Ports

The ports connected to PCs must be configured as access ports and assigned to the appropriate VLAN.

PC0 → VLAN 2

Switch0(config)# interface fastethernet0/1

Switch0(config-if)# switchport mode access

Switch0(config-if)# switchport access vlan 2

PC1 → VLAN 1

Switch0(config)# interface fastethernet0/2

Switch0(config-if)# switchport mode access

Switch0(config-if)# switchport access vlan 1

PC2 → VLAN 1

Switch0(config)# interface fastethernet0/3

Switch0(config-if)# switchport mode access

Switch0(config-if)# switchport access vlan 1

PC3 → VLAN 3

Switch0(config)# interface fastethernet0/4

Switch0(config-if)# switchport mode access

Switch0(config-if)# switchport access vlan 3

Similarly configure the ports connected to PC4, PC5, PC6 and PC7 on Switch1 according to their VLANs.

Configure Trunk Between Switches

The connection between **Switch0 and Switch1** carries multiple VLANs, so it must be configured as a trunk.

The inter-switch connection is through FastEthernet0/5:

Switch0

Switch0> enable

Switch0# configure terminal

Switch0(config)# interface fastethernet0/5

Switch0(config-if)# switchport mode trunk

Switch0(config-if)# switchport nonegotiate

Switch0(config-if)# switchport trunk allowed vlan 1-3

Switch1

Switch1> enable

Switch1# configure terminal

Switch1(config)# interface fastethernet0/5

Switch1(config-if)# switchport mode trunk

Switch1(config-if)# switchport nonegotiate

Switch1(config-if)# switchport trunk allowed vlan 1-3

Verification

Switch 0

```
Switch#show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/2, Fa0/3, Fa0/6, Fa0/7
                                                 Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                 Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                 Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                 Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                 Fa0/24, Gig0/1, Gig0/2
2    CSE                               active    Fa0/1
3    ECE                               active    Fa0/4
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/5       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/5       1-3

Port        Vlans allowed and active in management domain
Fa0/5       1,2,3

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/5       1,2,3
```

Switch1

```
Switch>enable
Switch#show vlan brief

VLAN Name                                Status    Ports
---- ----------------------------------- --------- -------------------------------
1    default                             active    Fa0/2, Fa0/3, Fa0/6, Fa0/7
                                                   Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                   Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                   Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                   Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                   Fa0/24, Gig0/1, Gig0/2
2    CSE                                 active    Fa0/1
3    ECE                                 active    Fa0/4
1002 fddi-default                        active
1003 token-ring-default                  active
1004 fddinet-default                     active
1005 trnet-default                       active
Switch#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/5       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/5       1-3

Port        Vlans allowed and active in management domain
Fa0/5       1,2,3

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/5       1,2,3
```

Output:

PDU List Window

Fire	Last Status	Source	Destination	Type
	Successful	PC1	PC6	ICMP
	Successful	PC3	PC7	ICMP
	Successful	PC0	PC4	ICMP
	Failed	PC0	PC3	ICMP
	Failed	PC1	PC3	ICMP
	Failed	PC1	PC4	ICMP

Justification

The VLAN configuration was verified using show vlan brief and show interfaces trunk. PCs belonging to the same VLAN were able to communicate through the trunk link between the two switches, while PCs belonging to different VLANs remained isolated. This confirms that the VLANs, access ports, and trunk configuration were configured successfully.